

Building an Interactive AWS Quiz Chatbot

Using Amazon Lex

An AI-powered chatbot designed to test learners' knowledge of AWS services.

Project: AWS Lex Chatbot Creation

Client: CloudLearners Inc.

Team Members

- Phenyo
- Tumelo
- Malaza
- Grace

What is a Chatbot?

A chatbot is a software program that simulates human conversation with users

Uses Natural Language Processing (NLP) to understand intent

Can understand user requests and respond automatically in real time

Examples



Customer Service Bots



Virtual Assistants



Educational Learning Bots

Our project focuses on building a chatbot using Amazon Lex.

Client Request

CloudLearners Inc. needed:

- An interactive chatbot for learners
- A knowledge quiz about Amazon S3
- Multiple choice questions
- Instant feedback for answers
- A simple and engaging learning experience



What is our main Goal ?

Help students learn AWS concepts through conversation.

Our Chatbot Solution

We built an interactive quiz chatbot using Amazon Lex.

Start Quiz



Users start with commands like "Start Quiz"

AWS Questions



The chatbot asks AWS-related questions

Multiple Choice



Users answer with A / B / C options

Instant Feedback



The bot provides immediate correct/incorrect responses

Why This Solution Is Valuable

Benefits for CloudLearners Inc.

 Automates learning assessments

 Provides interactive education

 Gives instant feedback to learners

 Can scale to thousands of users

 Reduces need for manual trainers

Key Impact: Makes AWS learning more engaging and accessible.

How the Chatbot Works

Core components used in Amazon Lex:



Bot

Main chatbot system



Intent

Defines what the user wants



Utterances

Example phrases users type



Slots

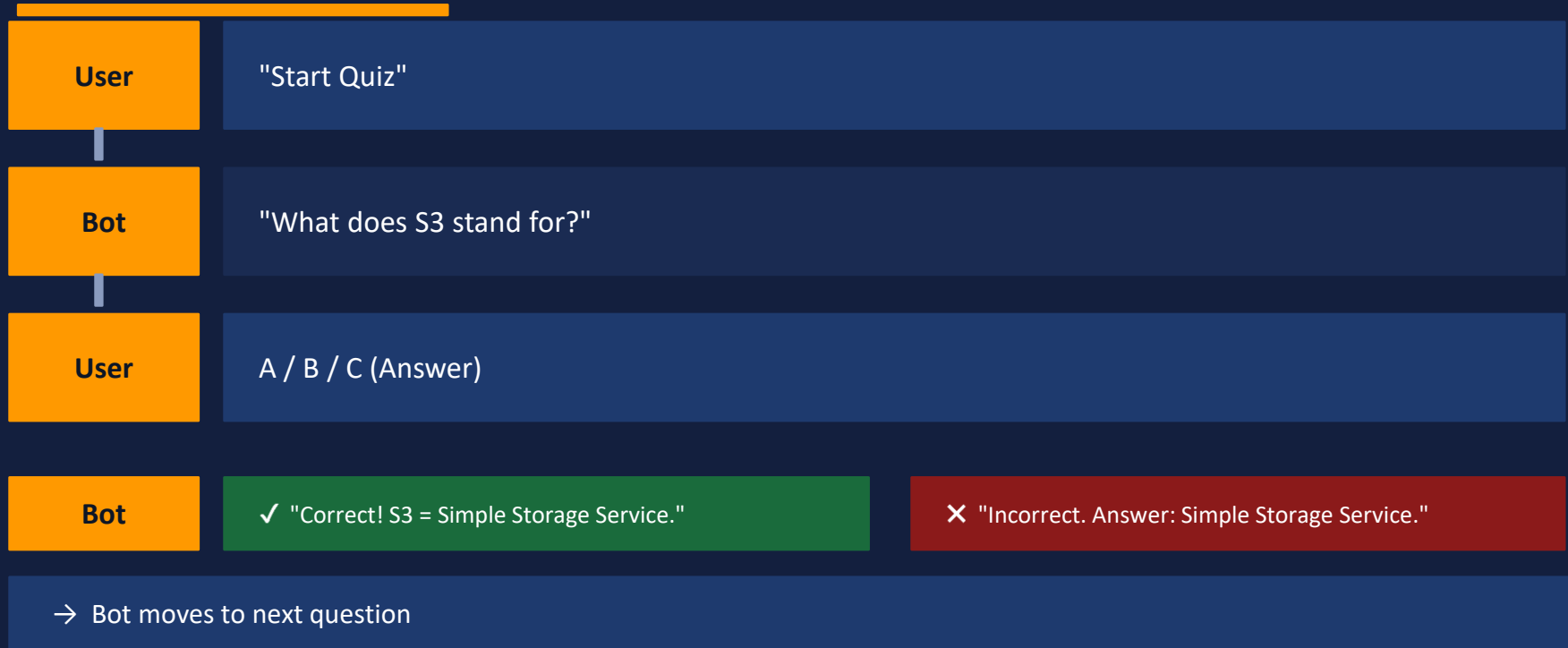
Capture user responses



Responses

What the bot replies with

Quiz Interaction Flow



Live Chatbot Demonstration

In this demo we will show:

- 1 Starting the chatbot
- 2 Triggering the quiz intent
- 3 Answering quiz questions
- 4 Bot responses for correct answers
- 5 Bot responses for incorrect answers

We will now demonstrate the chatbot working in Amazon Lex.

Challenges We Faced

1

Challenge 1 — Exact Slot Naming

Problem: Slot values had to match exactly — incorrect wording prevented the bot from working.

Solution: Standardized slot values and responses throughout the bot configuration.

2

Challenge 2 — Slot Type Confusion

Problem: Initially tried creating custom slot types which caused unexpected errors.

Solution: Discovered and used pre-configured Amazon Lex slot types instead.

Key Takeaways



Amazon Lex enables AI-powered conversational interfaces



Chatbots can enhance interactive learning platforms



Proper configuration of intents, slots, and utterances is critical

This project demonstrates how AWS AI services can be used to build scalable educational tools.

Questions?

Thank you for listening.

We welcome any questions.

Phenyo

Tumelo

Nhlanhla

Grace